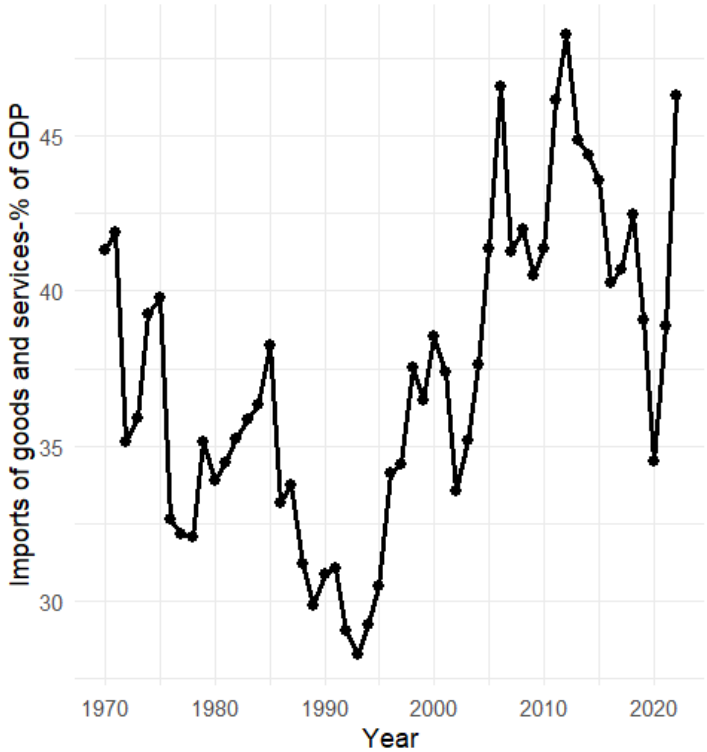

Figure 1. Graphical representation of the time series of the annual contribution of imports of goods and services to GDP (% of GDP).

The graph shows the time series referring to the annual share of imports of goods and services in GDP in Iceland. Since the time series is annual, seasonality is not present.

A trend is visible within several phases that can be roughly identified from the figure: from 1970 to the early 1990s, from the early 1990s to 2005, and the final phase after 2005. In line with the graph, the series shows a long-run transition (structural break) from a declining regime (1970s to 1990s) to a growing regime (after the 2000s), with shocks around 2010 and a clear shock around 2020.

The observed structural breaks and shocks may be a consequence of changes in the global economy during specific periods, or changes within Iceland that affected the share of imported goods and services in GDP. The clear shock around 2010 was probably related to the global crisis that began in 2008, with recession effects visible through 2009–2010, while the shock around 2020 was likely influenced by the COVID-19 pandemic (because of reduced imports of goods and especially reduced tourism).

To examine and identify the structural changes (breaks) more precisely, as suggested by the initial graph (Task 1), a structural break analysis was performed. This analysis was used so that, among possible ARIMA solutions, one model could also be estimated on the part of the data starting from the last structural break, and then compared with a model fitted on the full dataset.

As the number of breaks increases (up to five), RSS (residual sum of squares) decreases, but the number of model parameters increases. Therefore, the BIC criterion was used to determine the number of breaks, and based on it the number of structural breaks was estimated as 3 (the lowest BIC). A graphical display of the break selection is shown below.

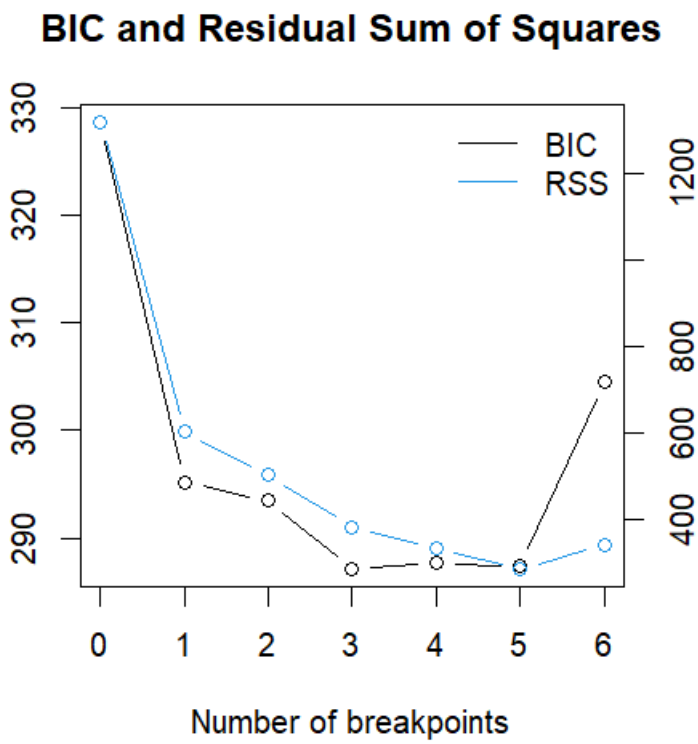

Figure. Graphical display of the number of breaks selected using the BIC criterion.

From the graph it can be seen that RSS decreases substantially when three structural breaks are introduced. The years in which the breaks occur are 1987, 1995, and 2004 (shown in Figure 4), with 95% confidence intervals (shown in Figure 3): [1986, 1990], [1994, 1996], and [2000, 2005].

```
Corresponding to breakdates:
  2.5 % breakpoints 97.5 %
1  1986        1987   1990
2  1994        1995   1996
3  2000        2004   2005
```

Figure. Confidence intervals of the assigned break dates.

```
Corresponding to breakdates:

m = 1                          2004
m = 2   1976                   2004
m = 3               1987 1995  2004
m = 4   1976        1987 1995  2004
m = 5   1976        1987 1995  2004 2015
m = 6   1976 1983   1990 1997  2004 2015

Fit:

m   0      1      2      3      4      5      6
RSS 1318.5  603.0  502.7  383.9  334.0  286.1  340.5
BIC  328.7  295.2  293.5  287.1  287.7  287.4  304.6
```

Figure. Identified years in which the breaks occur.

Display and comment on the ACF and PACF of the observed time series in levels.

ACF is the autocorrelation function, and each bar on the graph represents the correlation between x_t and x_(t-k), where k is the lag taking values from 0 to 20 (for lag 0 the autocorrelation equals 1). Each bar is obtained as corr(x_t, x_(t-k)), where the correlation is calculated for all pairs in the time series. The ACF bars decline as the lag increases because the correlation is measured with increasingly distant past values.

A large autocorrelation at small lags (from 1 to 8) shows that the values are similar from year to year, while the slow decline toward 0 indicates a possible trend. Since there is no sinusoidal oscillation, the time series is not cyclical.

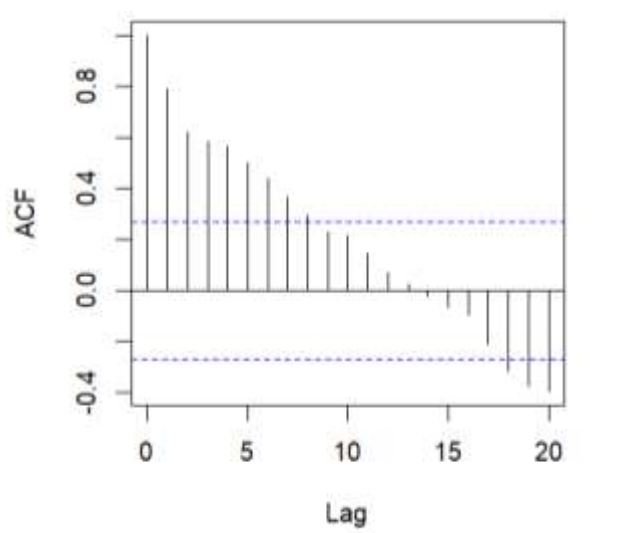

Figure 5. ACF of the observed time series in levels.

As an additional step, a three-year moving average was computed, which shows non-stationarity in the mean and, together with the ACF, suggests the possible presence of a trend.

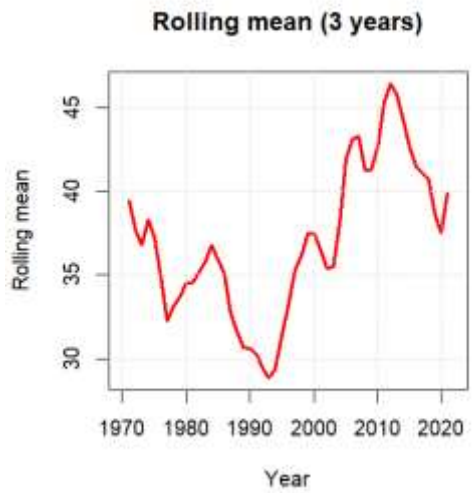

Figure. Three-year moving average of the time series.

PACF (partial autocorrelation) is the direct relationship between x_t and x_(t-k) after removing the intermediate lags, that is, corr(x_t, x_(t-k) | x_(t-1) ... x_(t-k)). The graph shows that lag 1 is high, meaning there is strong dependence on the previous year even after controlling for the others. This suggests that a strong AR(1) component may be present. However, to study the partial autocorrelation in more detail, differencing must be performed so that the series becomes stationary.

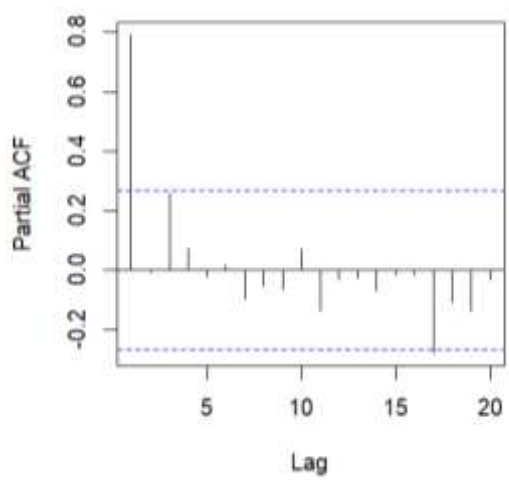

Figure. PACF of the observed time series in levels.

Examine and comment on the Ljung-Box Q test.

To test for autocorrelation, the Ljung-Box Q test is used, where:

H0: the ACF at all lags equals 0 (there is no autocorrelation), p1 = p2 = ... = pm.

H1: the ACF differs from 0 at least at one lag.

```
        Box-Ljung test

data:  x_Train
X-squared = 184.44, df = 20, p-value < 2.2e-16
```

Figure. Ljung-Box Q test for the time series in levels.

Since the p-value is < 2.2e-16, that is, smaller than 0.05, H0 is rejected and the alternative hypothesis that statistically significant autocorrelation exists is accepted. This is consistent with the ACF graph and indicates that the series in levels is highly autocorrelated and therefore not suitable for direct ARIMA modeling.

Test for the presence of a unit root using the Augmented Dickey-Fuller test and comment on the results.

To test stationarity through the presence of a unit root, the ADF test is used, where:

H0: the time series has a unit root (it is non-stationary).

H1: the time series does not have a unit root (it is stationary).

```
        Augmented Dickey-Fuller Test

data:  x_Train
Dickey-Fuller = -2.1695, Lag order = 3, p-value = 0.5063
alternative hypothesis: stationary
```

Figure. ADF test for the time series in levels.

p-value = 0.5063 -> we do not reject H0 -> the series is probably non-stationary in levels (also observed in the earlier graphs through trend / level change).

Display and comment on the graphical representation of the observed time series in the appropriate difference based on the ADF test, and compare it with the graph of the series in levels.

Since the ADF test indicates that the series is non-stationary, we difference it:

X'_t = X_t - X_(t-1).

After the first difference, we apply the ADF test again to check stationarity.

```
        Augmented Dickey-Fuller Test

data:  x_Traindiff1
Dickey-Fuller = -4.0186, Lag order = 3, p-value = 0.01555
alternative hypothesis: stationary
```

Figure. ADF test for the first-differenced time series.

p-value = 0.01555 -> we reject H0 -> the series is stationary, so additional differencing is not necessary.

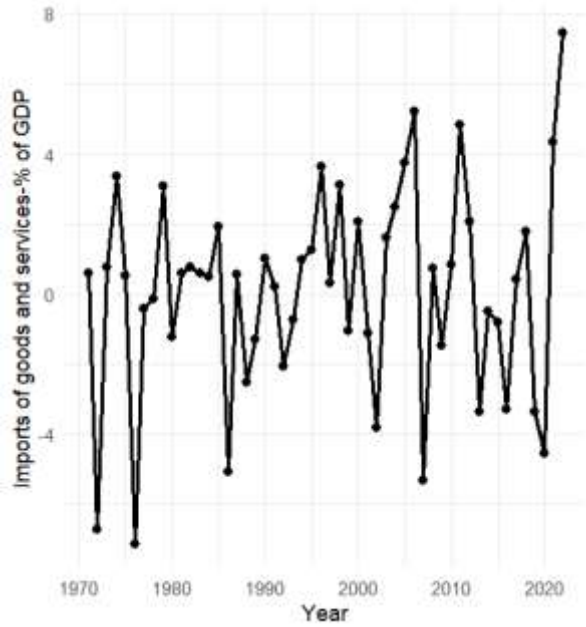

Figure. Graphical representation of the differenced time series.

In the new graph (Figure 11) of the differenced time series, there is no clearly visible trend, which indicates that the time series is stationary. In contrast, the original series in levels showed a long-term movement and structural changes.

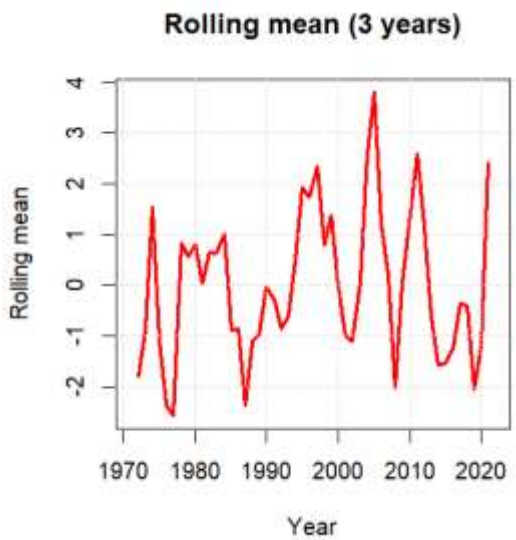

Figure. Three-year moving average of the differenced time series.

As an additional step, a three-year moving average was calculated for the differenced series (Figure 12), which now shows that the time series oscillates around 0, which is the expected behavior of a stationary series without trend.

Draw and comment on the ACF and PACF of the observed time series in the appropriate difference based on the ADF test.

Based on the ACF graph (Figure 13), the MA component (q) is most likely 0, because the bars are not statistically significant (all are within the blue bounds).

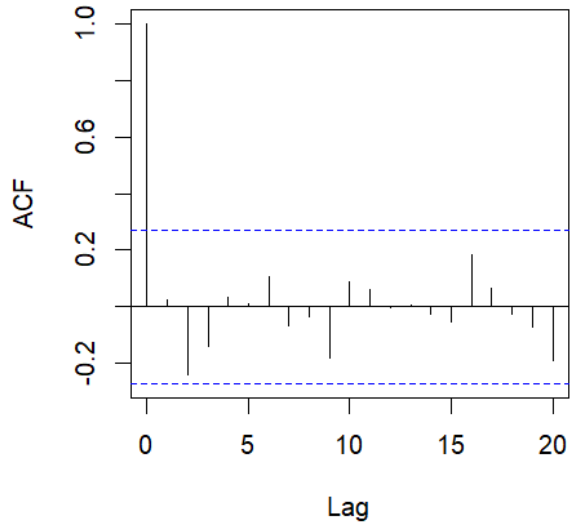

Figure 13. ACF of the differenced time series.

In the PACF graph of the stationary series (Figure 14), the AR component (p) is also most likely 0, because the bars are not statistically significant (all are within the blue bounds).

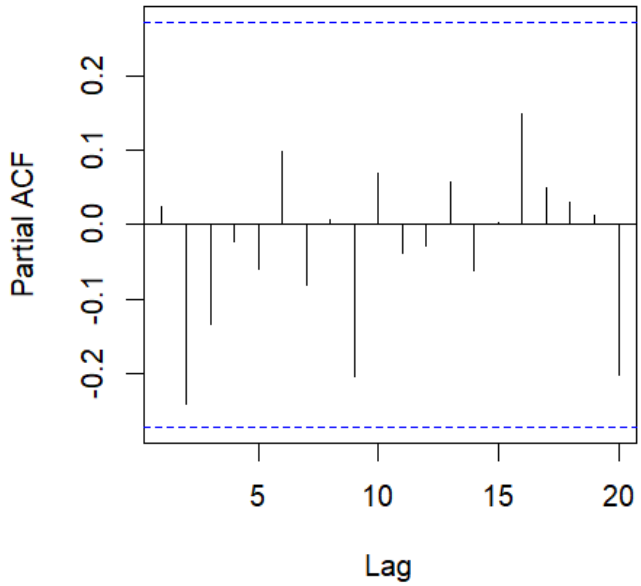

Figure. PACF of the differenced time series.

Based on the above, the time series will most likely be represented by an ARIMA(0,1,0) model. However, these are heuristics and it is necessary to examine the statistical significance of the model coefficients.

Examine and comment on the Ljung-Box Q test.

```
        Box-Ljung test

data:  x_Traindiff1
X-squared = 4.4104, df = 4, p-value = 0.3533
```

Figure. Ljung-Box Q test for the differenced time series.

The Ljung-Box Q test examines the autocorrelation of the time series:

H0: the ACF at all lags equals 0 (there is no autocorrelation): p1 = p2 = ... = pm.

H1: the ACF differs from 0 at least at one lag.

Since the p-value is > 0.05, we do not reject H0, which means that there is no evidence of autocorrelation. This is consistent with the observed properties of the ACF graph. The graph obtained after differencing is therefore suitable for further modeling.

Model the observed time series using an ARIMA model. Present 2–3 model solutions with the corresponding formulas and test the significance of the coefficients.

Based on the displayed graphs and tests, the most likely ARIMA model is a random walk, ARIMA(0,1,0), which means that p = 0 and q = 0, that is:

$X_t - X_{(t-1)} = c + \varepsilon_t$.

The drift is approximately 0.096, so $X_t - X_{(t-1)} = 0.096 + \varepsilon_t$. This means that on average the time series increases by about 0.096 units per year. Because of random noise, individual years may deviate from this average.

```
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
drift 0.096074   0.409944  0.2344   0.8147
```

Figure. z-test of coefficient significance for the ARIMA(0,1,0) model.

Test of coefficient significance: H0: c = 0 (there is no average trend / drift).

H1: $c \neq 0$ (drift exists).

Since the p-value is 0.8147 > 0.05, H0 is not rejected, meaning that the estimated coefficient is not statistically significant. This means the estimate is imprecise and that this is a random walk without drift.

If we consider a model with an MA(1) component, ARIMA(0,1,1):

$X_t = 0.052153\, X_{(t-1)} + 0.104140 + \varepsilon_t$.

```
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ma1   0.052153   0.199371  0.2616   0.7936
drift 0.104140   0.431735  0.2412   0.8094
```

Figure. z-test of coefficient significance for the ARIMA(0,0,1) model.

Since the p-value for the drift is 0.80947 > 0.05, and for the coefficient 0.7836 > 0.05, H0 is not rejected, that is, the obtained coefficients are not statistically significant. The same result was obtained for ARIMA(1,1,0).

```
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1    0.02614    0.14633  0.1786   0.8582
drift  0.10011    0.42124  0.2377   0.8121
```

Figure. z-test of coefficient significance for the ARIMA(1,1,0) model.

Select the appropriate solution according to the AIC criterion and comment on the results.

Since the estimated drift is not statistically significant (p > 0.05), the subsequent analysis was conducted on models without drift. Comparing ARIMA(0,1,0), ARIMA(0,1,1), and ARIMA(1,1,0), the best solution according to AIC is ARIMA(0,1,0).

```
> aicModel<-c(AIC(m1), AIC(m2) ,AIC(m3))
> aicModel
[1] 262.3487 264.2869 264.3185
```

Figure. Selection of the solution based on the AIC criterion.

Use the auto.arima() function to find the best model solution — the final model. Present the final model with the appropriate formula and test it.

```
Series: x_Train
ARIMA(0,1,0)

sigma^2 = 8.748:  log likelihood = -130.17
AIC=262.35   AICc=262.43   BIC=264.3

Training set error measures:
                     ME     RMSE      MAE         MPE     MAPE      MASE      ACF1
Training set 0.09504032 2.929676 2.216658 -0.07730952 5.910539 0.9814772 0.0233857
```

Figure 20. Selection of the solution based on the auto.arima() function.

1.1 The final model obtained by the auto.arima() function (Figure 20) is ARIMA(0,1,0), a random walk:

$X_t - X_{(t-1)} = \varepsilon_t$.

1.2 Since structural breaks were observed at the beginning, for the purpose of comparing forecasting performance in later predictions, in addition to the model above an auto.arima() model fitted only on the data from the last structural break onward (from 2004) will also be used.

```
ARIMA(0,0,1) with non-zero mean

Coefficients:
         ma1     mean
      0.7381  42.2137
s.e.  0.1680   1.0610

sigma^2 = 8.229:  log likelihood = -46.32
AIC=98.64   AICc=100.24   BIC=101.47
```

Figure. Model obtained by auto.arima() for part of the dataset (from 2004 onward).

Since this is ARIMA(0,0,1) with a nonzero mean, it is actually an MA(1) model with an intercept (the average level around which the series oscillates):

$X_t = 42.213 + \varepsilon_t + 0.738\, \varepsilon_{(t-1)}$.

A Wald (z) test of coefficient significance was carried out, where H0 states that the coefficient is not statistically significant. Since the obtained p-values are < 0.05, H0 is rejected and the coefficients are statistically significant.

```
z test of coefficients:

          Estimate Std. Error z value  Pr(>|z|)
ma1        0.73806    0.16803  4.3923 1.121e-05 ***
intercept 42.21373    1.06095 39.7885 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure. z-test of coefficient significance for the MA(1) model fitted on data from 2004 onward.

This means that the current value equals the average nonzero level of the series (42.213), increased by today’s random shock and part of the previous random shock ($0.738\, \varepsilon_{(t-1)}$).

Display the residuals graphically: series plot, ACF, histogram.

For ARIMA(0,1,0), the residual ACF does not indicate autocorrelation, while the residuals visually deviate from the normal distribution.

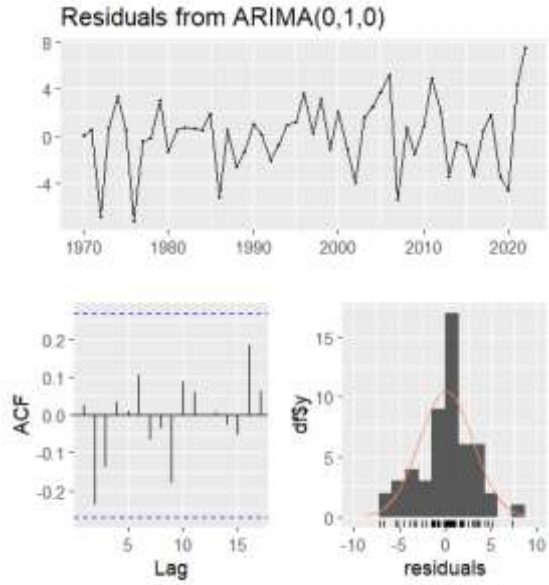

Figure. Model plot, residual ACF, and residual histogram for ARIMA(0,1,0).

To test whether the residuals come from a normal distribution, the Shapiro–Wilk test was used (suitable for smaller samples): H0: the residuals are normally distributed; H1: the residuals are not normally distributed.

```
> shapiro.test(residuals(fit_2004))

        Shapiro-Wilk normality test

data:  residuals(fit_2004)
W = 0.9417, p-value = 0.2829
```

Figure. Shapiro–Wilk test for residual normality.

For MA(1):

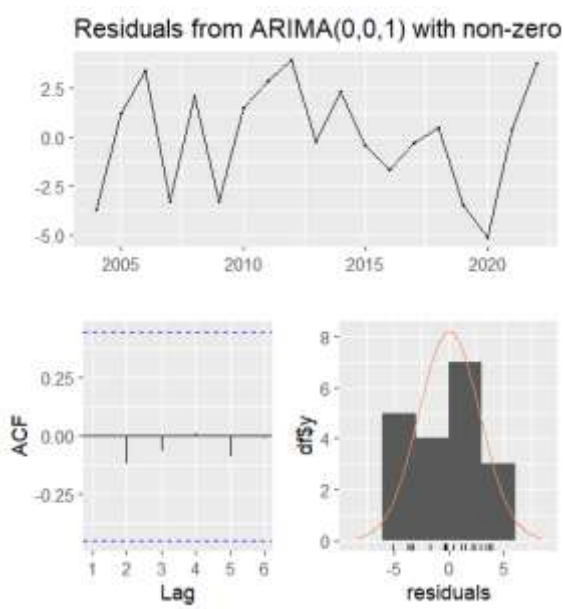

Figure. Model plot, residual ACF, and residual histogram for ARIMA(0,0,1).

Based on the graph (Figure 25), the conclusions are similar to the previous case: visually there is no autocorrelation (ACF), while the residuals are not normally distributed. The Shapiro–Wilk test confirms this.

Test for the presence of autocorrelation in the final model.

H0: there is no significant autocorrelation. H1: there is significant autocorrelation.

ARIMA(0,1,0).

p > 0.05 -> we do not reject H0 -> there is no autocorrelation in the final model.

The residuals behave like white noise, which is a good sign for model adequacy. Model df indicates that p and q of the model are 0, while total lags used indicates that autocorrelation is checked up to lag 10.

```
        Ljung-Box test

data:  Residuals from ARIMA(0,1,0)
Q* = 8.1884, df = 10, p-value = 0.6104

Model df: 0.   Total lags used: 10
```

Figure. Ljung-Box test for autocorrelation in the final ARIMA(0,1,0) model.

MA(1).

```
> checkresiduals(fit_2004)

        Ljung-Box test

data:  Residuals from ARIMA(0,0,1) with non-zero mean
Q* = 0.45373, df = 3, p-value = 0.9289

Model df: 1.   Total lags used: 4
```

Figure. Ljung-Box test for autocorrelation in the final ARIMA(0,0,1) model on part of the data.

p > 0.05 -> we do not reject H0 -> there is no autocorrelation in the final model.

The residuals behave like white noise, which is a good sign for model adequacy. Model df indicates that p + q of the model is 1 (an MA component is present), while total lags used indicates that autocorrelation is checked up to lag 4.

Test for the presence of heteroskedasticity in the final model.

ARIMA(0,1,0)

```
> e <- residuals(fit) #reziduali modela
> ArchTest(e)

        ARCH LM-test; Null hypothesis: no ARCH effects

data:  e
Chi-squared = 10.472, df = 12, p-value = 0.5746
```

Figure. ARCH LM test for heteroskedasticity of residuals in the final ARIMA(0,1,0) model.

H0: homoskedasticity (if p > 0.05, we accept H0).

H1: heteroskedasticity.

MA(1).

```
        ARCH LM-test; Null hypothesis: no ARCH effects

data:  e1
Chi-squared = 7, df = 12, p-value = 0.8576
```

Figure. ARCH LM test for heteroskedasticity of residuals in the final ARIMA(0,0,1) model on part of the data.

For this model as well, the p-value is > 0.05, so the residual variance is homoskedastic.

Forecast the observed time series for the period 2023–2033.

To forecast for the specified period, for ARIMA(0,1,0) all available data up to 2022 are used, while for ARIMA(0,0,1) the data from 2004 onward are used.

Provide a tabular display of the forecasts with the corresponding confidence intervals.

Because the Shapiro–Wilk test showed p < 0.05, bootstrap prediction intervals were also calculated, since they do not require the assumption of normally distributed residuals. However, standard intervals are also shown for comparison.

1.1 ARIMA(0,1,0).

	Point Forecast	Lo 95	Hi 95	Lo 99	Hi 99
2023	46.3071	40.51009	52.10411	38.68854	53.92566
2024	46.3071	38.10889	54.50531	35.53283	57.08137
2025	46.3071	36.26638	56.34781	33.11136	59.50283
2026	46.3071	34.71308	57.90112	31.06998	61.54422
2027	46.3071	33.34459	59.26961	29.27148	63.34272
2028	46.3071	32.10738	60.50681	27.64551	64.96869
2029	46.3071	30.96965	61.64454	26.15028	66.46392
2030	46.3071	29.91068	62.70352	24.75855	67.85564
2031	46.3071	28.91607	63.69813	23.45141	69.16278
2032	46.3071	27.97535	64.63885	22.21509	70.39911
2033	46.3071	27.08059	65.53360	21.03919	71.57501

Figure. Tabular display of forecasts from the ARIMA(0,1,0) model with the corresponding confidence intervals.

MA(1).

	Point Forecast	Lo 95	Hi 95	Lo 99	Hi 99
2023	44.99702	39.37460	50.61944	37.60791	52.38614
2024	42.21373	35.22578	49.20168	33.03001	51.39746
2025	42.21373	35.22578	49.20168	33.03001	51.39746
2026	42.21373	35.22578	49.20168	33.03001	51.39746
2027	42.21373	35.22578	49.20168	33.03001	51.39746
2028	42.21373	35.22578	49.20168	33.03001	51.39746
2029	42.21373	35.22578	49.20168	33.03001	51.39746
2030	42.21373	35.22578	49.20168	33.03001	51.39746
2031	42.21373	35.22578	49.20168	33.03001	51.39746
2032	42.21373	35.22578	49.20168	33.03001	51.39746
2033	42.21373	35.22578	49.20168	33.03001	51.39746

Figure. Tabular display of forecasts from the MA(1) model with the corresponding confidence intervals.

Provide a graphical display of the forecasts with the corresponding confidence intervals.

1.1

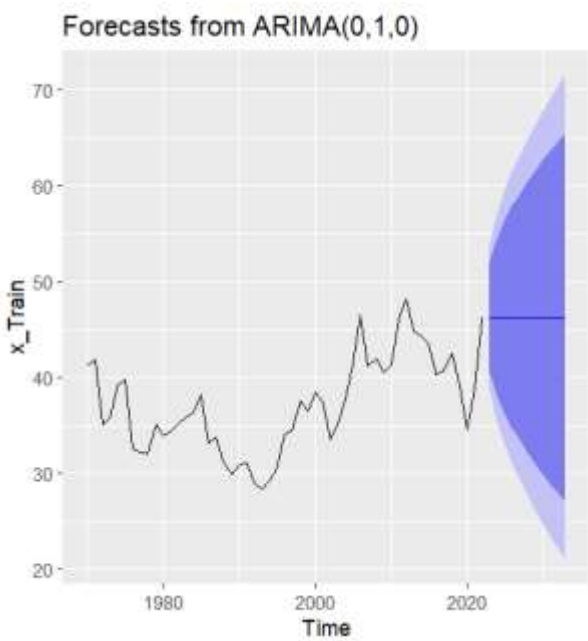

Figure. Graphical display of the forecasts using the ARIMA(0,1,0) model with the corresponding confidence intervals.

1.2

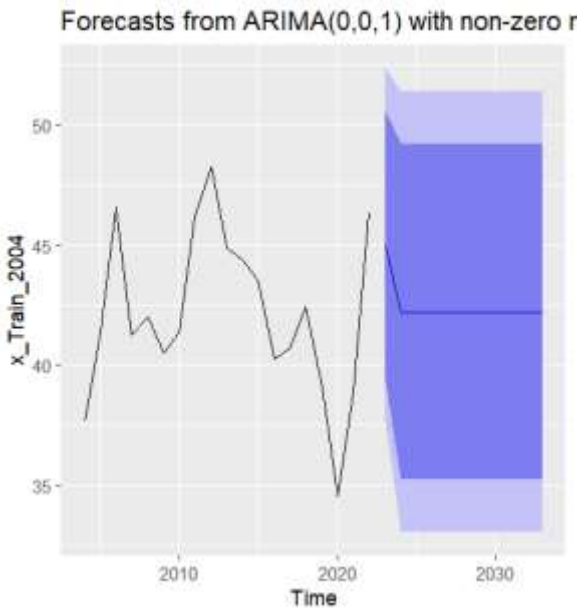

Figure. Graphical display of the forecasts using the ARIMA(0,0,1) model with the corresponding confidence intervals.

For the year 2023, compare the obtained results with the actual value and provide comments.

For the comparison, the available data for both 2023 and 2024 were used for both models, so the forecast was given not only for 2023 but also for 2024.

```
> accuracy(fc, x_Test)
                     ME     RMSE      MAE         MPE     MAPE      MASE       ACF1 Theil's U
Training set  0.09504032 2.929676 2.216658 -0.07730952 5.910539 0.9814772  0.0233857        NA
Test set     -2.91997036 2.922778 2.919970 -6.73096846 6.730968 1.2928851 -0.5000000  10.89909
> accuracy(fc_2004, x_Test)
                     ME     RMSE      MAE        MPE     MAPE      MASE         ACF1 Theil's U
Training set  0.03334605 2.713453 2.287859 -0.3633249 5.513422 0.7603584 -0.004775129        NA
Test set     -0.21825101 1.535316 1.519724 -0.5133762 3.504223 0.5050727 -0.500000000  5.080743
```

Figure. Comparison of the results of the ARIMA(0,1,0) model on the full dataset and the ARIMA(0,0,1) model on a subset of the data.

```
> AIC   = c(AIC(fit), AIC(fit_2004))
> AIC
[1] 262.34868  98.63898
```

Figure. Comparison using the AIC criterion for the ARIMA(0,1,0) model on the full dataset and the ARIMA(0,0,1) model on a subset of the data.

Comparing these two models (Figure 34 and Figure 36), better performance is observed for the MA(1) model (lower RMSE and MAE). The AIC is also substantially lower for the MA(1) model, but the AIC values are not strictly comparable because the models are fitted on different datasets.

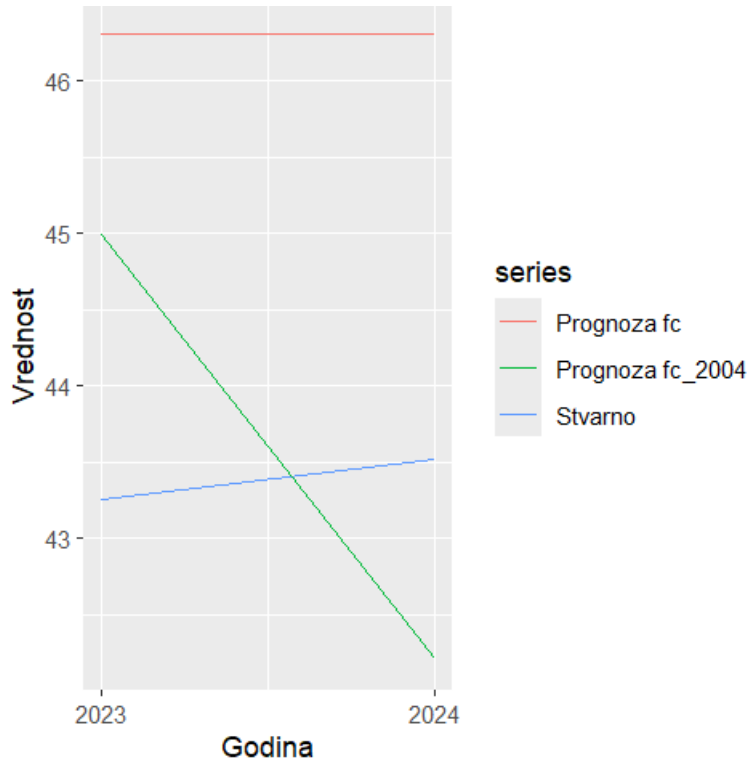

Figure. Graphical display of the predicted values of the ARIMA(0,1,0) and ARIMA(0,0,1) models.

Although the selected ARIMA models were chosen based on residual diagnostics, they do not track the time-series movement well in the test period. ARIMA(0,1,0) produces a forecast close to the previous value, while the MA(1) model on the post-break data better follows the newer regime but ignores the earlier part of the series.

This means that the data after the last detected break are more relevant for forecasting, but the problem is that the pre-break data are completely ignored in the ARIMA(0,0,1) case. Therefore, an ARIMAX model with a dummy variable for the structural break was introduced.

The general form of the model is: $X_t = \beta 0 + \beta 1\ D_t + U_t$. The term $\beta 0 + \beta 1\ D_t$ is the regression part of the model used to remove the effect of the break, while U_t is the residual component that is modeled by an ARIMA process.

The statistical significance of the obtained model coefficients was examined (Figure 37), and all coefficients were found to be statistically significant.

```
z test of coefficients:

          Estimate Std. Error z value  Pr(>|z|)
ar1        0.64156    0.12313  5.2105 1.883e-07 ***
intercept 35.36922    1.30214 27.1624 < 2.2e-16 ***
xreg       6.20725    1.97922  3.1362  0.001711 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 37. Testing the significance of the coefficients for forecasting using the ARIMAX model.

The form of the model obtained on the given data, after testing coefficient significance, is:

$X_t = 35.369 + 6.20725 D_t + U_t$, where $U_t = 0.64156 U_{(t-1)} + \varepsilon_t$.

Here the residual U_t is represented by an AR(1) model, and ε_t represents white noise. The Ljung-Box test showed that the residuals are not autocorrelated, the t-test showed that the mean is close to 0, and the ARCH test showed that there is no heteroskedasticity.

```
> ArchTest(residuals(fit_break), lags=5)

        ARCH LM-test; Null hypothesis: no ARCH effects

data:  residuals(fit_break)
Chi-squared = 4.2142, df = 5, p-value = 0.519
```

Figure. ARCH test of the ARIMAX residuals.

```
> t.test(residuals(fit_break))     # H0: mean=0

        One Sample t-test

data:  residuals(fit_break)
t = -0.2598, df = 52, p-value = 0.796
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.8552181  0.6591532
sample estimates:
  mean of x
-0.09803248
```

Figure. t-test of the ARIMAX residuals.

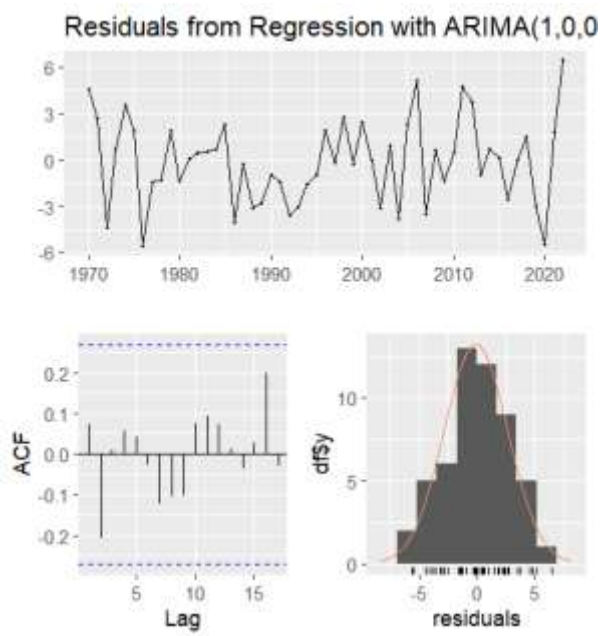

Figure. Model plot, residual ACF, and residual histogram for the ARIMAX model.

For comparison by the AIC criterion, ARIMAX and ARIMA(0,1,0) are compared because they are fitted on the same dataset. ARIMA(0,0,1) is excluded from this analysis because it is not fitted on an equal dataset.

```
aicc_tab
       fc break_dummy
 262.4287    265.9470
```

Figure 42. Comparison of ARIMAX and ARIMA(0,1,0) by the AICc criterion.

In the final forecasting comparison, the results are compared with the previous models, and ARIMAX is the model with the smallest forecasting errors.

```
                  RMSE      MAE     MAPE
fc           2.9227780 2.919970 6.730968
fc_2004      1.5353161 1.519724 3.504223
break_dummy  0.9563065 0.680386 1.572761
```

Figure 43. Comparison of ARIMAX with the previous models in terms of forecasting accuracy for 2023 and 2024.

In the initial check, a shorter test period (two years) was used, so to reduce the influence of the small number of points, the test period was extended (from 2016 onward). ARIMAX again gave better prediction accuracy in this broader test.

```
> rbind(ARIMA_010 = acc_rw, ARIMAX_dummy = acc_x)
                 RMSE      MAE     MAPE
ARIMA_010    4.103319 3.169773 8.242186
ARIMAX_dummy 3.469456 2.905480 7.344590
>
```

Figure 44. Comparison of ARIMAX with ARIMA(0,1,0) in terms of forecasting accuracy on the test set from 2016 onward.

Conclusion and discussion.

In this paper, graphical tools (ACF, PACF, and the time-series plot) were used to compare the time series in levels and its differenced version, and to assess the main properties of the series, including the presence of trend, autocorrelation, stationarity, extreme values, and structural breaks. After that, these assumptions were checked using the Ljung-Box test (for autocorrelation) and the ADF test (for stationarity).

When constructing the ARIMA models, several candidate models were estimated and the statistical significance of their coefficients was examined. The final models were then obtained using the auto.arima() function. Because structural breaks were observed, caused by changes in economic regimes that indirectly affect the percentage share of imports of goods and services in GDP, a model obtained with auto.arima() on only the data from the most recent regime was also used for comparison. The resulting models were:

Model fitted on part of the data: ARIMA(0,0,1): X(t) = 42.213 + e(t) + 0.738 e(t-1).

Model fitted on the full dataset: ARIMA(0,1,0): X(t) - X(t-1) = e(t).

Using these models for forecasting, it was found that the model fitted only on the data after the last structural break achieved better results according to AC, MSE, and RMSE, which indicates that structural

breaks should be incorporated into the model. However, a drawback of working only with part of the data was also observed, so compared with these two models, the following model stands out:

ARIMAX: $X(t) = 35.369 + 6.20725\ D(t) + U(t)$, where $U(t) = 0.64156\ U(t-1) + e(t)$.

This model uses the full time series, while also accounting for the regime change through a dummy variable for the structural break. It therefore represents a better compromise between using all available data and taking into account the change in the behavior of the series after the structural break. In the forecasting comparison, ARIMAX achieved the smallest forecasting errors, and when the test period was extended from 2016 onward, it still retained lower RMSE, MAE, and MAPE values, indicating that its advantage is not simply a consequence of a short test period.

A possible continuation of this analysis would be to perform validation through cross-validation in order to further verify the stability of the chosen model. In addition, ARIMAX could be improved further by introducing real economic explanatory variables, such as inflation, GDP growth, or the exchange rate, instead of relying only on a dummy variable for the structural break.